Development of the Monash True-Triaxial Hopkinson Bar System and an Integrated Digital Twin for Multiaxial Dynamic Rock Characterisation

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Abstract

Conventional split-Hopkinson pressure bar (SHPB) testing characterises rock under one-dimensional dynamic loading, yet rock in deep mining, tunnelling and rock blasting fails under *multiaxial* stress states at high strain rate. This paper presents the development of the Monash triaxial Hopkinson bar (Tri-HB) — a gas-gun-driven system of three independent pairs of orthogonal square bars with hydraulic confinement on all three axes, enabling true-triaxial ($\sigma_2 \neq \sigma_3$) static–dynamic coupled loading — and an integrated *digital twin*, the Virtual Tri-HB, that unifies test design, three-wave data reduction, stress-path interpretation and damage modelling in a single six-step workflow. We report four advances embedded in the digital twin: (i) a unified treatment of four loading families and six physics modes spanning uniaxial SHPB to electromagnetic programmable multiaxial loading; (ii) a true-triaxial, Lode-angle-dependent and rate-dependent failure envelope that captures the intermediate principal stress effect with the correct sign; (iii) a thermodynamically consistent damage-energy partition in which the first law closes by construction, $W_{\text{input}} = W_{\text{el}} + W_{\text{diss}}$; and (iv) an orthotropic damage tensor whose components are driven by the directional tensile strain, reproducing the directional (axial-splitting) fracture of rock blasting. Because the three orthogonal bars measure direction-resolved stress, strain and wave speed, each component of the damage tensor can in principle be calibrated independently — a capability a uniaxial SHPB cannot provide. We close with a calibrate-then-blind-predict validation roadmap that positions the Tri-HB and its digital twin for quantitative, multiaxial dynamic constitutive modelling of rock.

Keywords: triaxial Hopkinson bar; dynamic rock mechanics; true-triaxial loading; intermediate principal stress; anisotropic damage; rock blasting; digital twin.

1 Introduction

The dynamic mechanical response of rock governs the safety and efficiency of deep underground excavation, rockburst mitigation and rock blasting, where loading rates of 10^1–$10^3\,\mathrm{s}^{-1}$ coincide with confining stresses of tens to hundreds of megapascals (Zhang and Zhao, 2014; Xia and Yao, 2015). Capturing this response in the laboratory has driven more than a century of development of the Hopkinson bar, yet the classical apparatus remains one-dimensional. This section traces that development, examines how confinement has conventionally been added, and motivates a true-triaxial alternative.

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1.1 From the Hopkinson bar to the split-Hopkinson pressure bar

The technique originates with Bertram Hopkinson, who in 1914 used a long elastic bar terminated by a momentum trap to record the pressure–time history produced by explosive detonation and bullet impact (Hopkinson, 1914) — the original *Hopkinson pressure bar*. Davies (1948) replaced the mechanical momentum measurement with electrical (condenser) sensing of bar displacement and gave the first rigorous treatment of geometric wave dispersion. Kolsky (1949) then split the bar in two and placed a thin specimen between an incident and a transmission bar: the incident, reflected and transmitted pulses recovered from bar gauges yield the specimen stress, strain rate and strain, defining the split-Hopkinson pressure bar (SHPB), or Kolsky bar, still in use today. Subsequent maturation added resistance strain-gauge instrumentation, Pochhammer–Chree dispersion correction, and pulse shaping to enforce a constant strain rate and stress equilibrium in brittle specimens (Frew et al., 2002); the method was extended to tension, torsion and elevated temperature, and codified for rock through suggested methods and reviews (Zhou et al., 2012; Zhang and Zhao, 2014; Xia and Yao, 2015). Throughout, however, the loading remained *one-dimensional*: a single axial pulse on an unconfined specimen.

1.2 Confined and triaxial-chamber dynamic testing

Because rock in situ is confined, dynamic testing under lateral pressure followed soon after. Two strategies became standard. In *passive* confinement a rigid metal sleeve or shrink-fit ring surrounds the specimen and supplies a reactive radial constraint that rises with the axial load but is not independently controlled (Christensen et al., 1972). In *active* confinement the SHPB specimen is enclosed in a pressure cell and a hydraulic or pneumatic medium imposes a prescribed confining pressure before the dynamic axial pulse arrives (Lindholm et al., 1974). These confined-SHPB methods established the pressure dependence and dynamic increase of rock strength and remain widely used; they correspond to the confinement-chamber mode (Mode 2) of the present workflow. They share, however, three intrinsic limitations. First, the confinement is *axisymmetric*, $\sigma_2 = \sigma_3$, so the *intermediate principal stress effect* — shown by quasi-static true-triaxial testing to be first-order for rock strength (Mogi, 1971) — cannot be accessed. Second, the confinement is quasi-static and pre-applied; the lateral stress can be neither pulsed nor timed relative to the axial event. Third, the specimen is enclosed, so the lateral stress and strain are inferred through the cell rather than measured directly, and the lateral dynamic (Poisson-driven) response is not resolved.

From a boundary-value standpoint, the confining medium acts as a low-impedance *traction* boundary rather than a displacement constraint. On the wetted specimen surface S with outward normal $\mathbf{n}$, an inviscid fluid transmits only a normal pressure and couples to the wall acceleration through

$$\sigma_{ij} n_j = -p(t)\, n_i, \qquad \frac{\partial p}{\partial n} = -\rho_f\, \ddot{u}_n, \tag{1}$$

the first a Neumann (traction) condition that leaves the lateral displacement free, the second the fluid–structure coupling that fixes any radiated energy. The acoustic mismatch with the rock, $R_{\text{lat}} = (Z_f - Z_s)/(Z_f + Z_s)$ with $Z = \rho c$, is severe — $Z_f/Z_s \sim 10^{-1}$ for liquids and $\lesssim 10^{-3}$ for gases — so the surface behaves almost as a pressure-release boundary and radiates little, while the chamber pressure stays effectively quasi-static, $p \approx p_c$, over the $\sim 100\,\mu$s event. The confinement is thus a clean axisymmetric pressure ($\sigma_2 = \sigma_3 = p_c$) that neither suppresses lateral expansion nor yields a measurable lateral signal.

These chamber effects can be mitigated, and the relations above prescribe how. The dynamic pressure rise, $\Delta p \approx K_{\text{eff}}(V_s/V_f)\, 2\nu\varepsilon_{\text{ax}}$, is held well below p_c by a large reservoir ($V_f \gg V_s$) or a soft medium — a gas, with $K_{\text{eff}} = \gamma p_c$, is about two orders of magnitude better than hydraulic oil. Radiation into the medium is suppressed by smooth, long-wavelength loading ($ka \ll 1$, the same condition that controls bar dispersion); wall reflections are excluded from the measurement window t_w by sizing the chamber radius $d > c_f t_w/2$; and parasitic load paths are removed with a thin, soft sealing jacket ($A_j E_j \ll A_s E_s$) and non-load-bearing seals. Each remedy, however, only *mitigates* a low-impedance, enclosed boundary, and none recovers the lateral signal — the gap the next section closes by replacing the chamber with matched-impedance bars.

1.3 The case for a true-triaxial Hopkinson bar

The Monash triaxial Hopkinson bar (Tri-HB) removes the confining chamber altogether and instead loads a cubic specimen through *three independent pairs of orthogonal Hopkinson bars*, with hydraulic cylinders applying independent quasi-static pre-stress on each axis and a gas gun superposing the dynamic pulse on the X axis. Each limitation above becomes a capability. (i) *True-triaxial* states, $\sigma_1 > \sigma_2 > \sigma_3$, are imposed by setting the three pre-stresses independently, so the intermediate-stress and Lode-angle dependence can be probed dynamically. (ii) *Static–dynamic coupling* reproduces the in-situ condition of a pre-stressed rock mass struck by a dynamic disturbance, as in blasting or rockburst. (iii) The lateral response is *measured directly*: each transverse axis carries its own output bars and gauges (six gauge sets in total), so σ_y, σ_z and the directional fracture are resolved rather than inferred. (iv) With programmable electromagnetic topologies the per-axis pulses can be timed independently, enabling loading-path and sequence studies a chamber cannot provide.

These hardware capabilities become quantitative only if the interpretation keeps pace. Reducing a multiaxial dynamic test to a constitutive statement requires a consistent three-wave analysis on each axis, a failure surface that depends on all three stress invariants, a thermodynamically admissible energy balance, and a damage description that is *directional* — because blasting produces oriented cracks rather than isotropic softening. These ingredients are rarely assembled in one auditable workflow.

1.4 Scope and outline

This paper presents the Tri-HB system together with an integrated digital twin that couples the hardware data-reduction equations to such a constitutive framework. Section 2 describes the hardware and its four loading families; Section 3 the one-dimensional wave propagation, the per-axis three-wave reduction, and the wave-interaction regime map; and Section 4 the integrated digital twin. Section 5 presents the three constitutive advances — the true-triaxial rate-dependent envelope, the energy-consistent dissipation, and the orthotropic damage tensor. Section 6 gives the validation roadmap, and Section 7 concludes. All figures are generated by the Virtual Tri-HB workspace from prescribed-pulse reference cases; the present contribution is the system and its modelling framework, with experimental calibration identified as the immediate next step.

2 The Monash Tri-HB system

2.1 Hardware configuration

The Tri-HB (Fig. 1) couples a gas-gun dynamic loading train on the X axis with three independent pairs of orthogonal square bars and hydraulic confinement on all three axes. The dynamic train comprises a cylindrical striker (∅40 mm, length 0.5 m), an incident bar (2.5 m), a transmission bar (2 m), an absorption bar and a momentum trap, all of 42CrMo steel (Table 1). The lateral Y and Z axes each carry two 2 m square output bars. A cubic specimen is sandwiched at the centre, producing six rock–bar interfaces with no bar-to-bar contact; support wheels constrain each bar to its own axis. Three hydraulic cylinders (capacity up to 100 MPa) apply independent quasi-static pre-stress on X, Y and Z, while the gas gun (impact velocity up to 50 m/s) superposes the dynamic pulse on X. The result is genuine static–dynamic coupled, true-triaxial loading: the pre-stress sets a multiaxial initial state and the striker drives the high-rate event. Six sets of

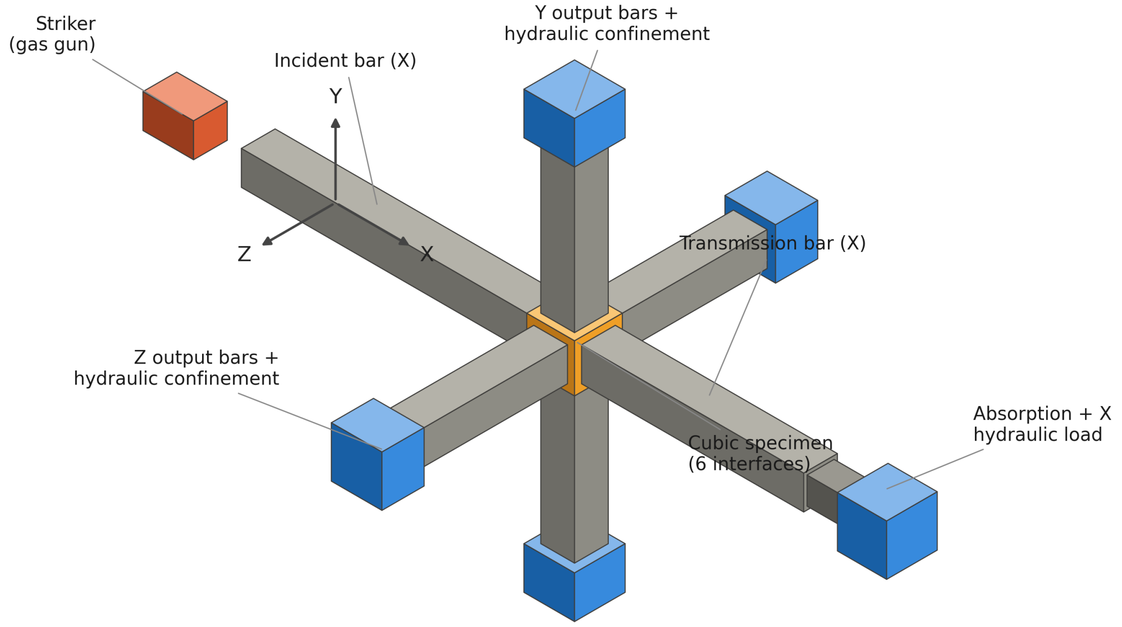


Figure 1: Isometric schematic of the Monash Tri-HB. The gas gun drives the X incident/transmission train; two output-bar pairs on Y and Z and a hydraulic load cylinder on X apply independent confinement, giving static–dynamic coupled true-triaxial loading on the central cubic specimen.

strain gauges (sampling at 1 MHz) on the incident, transmission and four output bars record the incident, reflected, transmitted and lateral-output strains, $\varepsilon_{\mathrm{in}}$, $\varepsilon_{\mathrm{re}}$, $\varepsilon_{\mathrm{tr}}$, $\varepsilon_{y1,y2}$, $\varepsilon_{z1,z2}$.

2.2 Four loading families

The digital twin exposes the system as four top-level loading families, the last of which branches into three electromagnetic (EM) topologies, giving six physics modes in total (Table 2). This taxonomy spans the historical uniaxial SHPB (Mode 1), passively or actively confined SHPB (Mode 2), the gas-gun Tri-HB static–dynamic configuration (Mode 3), and EM programmable loading (Modes 4–6) in which the pulse on each axis is prescribed independently. Modes 3 and 5 realise the true-triaxial capability that distinguishes the system.

3 Per-axis data reduction and wave interaction

3.1 One-dimensional wave propagation and dispersion

Each pressure bar is a long, slender rod in which, for wavelengths much greater than the bar diameter, the axial displacement $u(x,t)$ obeys the one-dimensional elastic wave equation

$$\frac{\partial^2 u}{\partial t^2} = C_0^2 \frac{\partial^2 u}{\partial x^2}, \qquad C_0 = \sqrt{E_b/\rho_b}, \tag{2}$$

whose general d'Alembert solution $u = f(x - C_0 t) + g(x + C_0 t)$ superposes forward- and backward-travelling parts. For a wave travelling in a single direction the axial strain and particle velocity

Table 1: Representative Monash Tri-HB bar and loading parameters (42CrMo steel).

Quantity	Symbol	Value
Bar density	ρ_b	$7850\,\mathrm{kg/m^3}$
Bar Young's modulus	E_b	$210\,\mathrm{GPa}$
Bar wave speed	C_0	$5200\,\mathrm{m/s}$
Bar yield strength	σ_y	$930\,\mathrm{MPa}$
Striker (diameter, length)	$\varnothing$, L_s	$40\,\mathrm{mm}$, $0.5\,\mathrm{m}$
Square-bar cross-section	A_b	$50\,\mathrm{mm} \times 50\,\mathrm{mm}$
Incident / transmission / output bar	L	2.5 / 2.0 / $2.0\,\mathrm{m}$
Maximum impact velocity	v	$50\,\mathrm{m/s}$
Hydraulic confinement capacity	p_c	$100\,\mathrm{MPa}$
Gauge sampling rate	—	$1\,\mathrm{MHz}$

Table 2: Loading families and the six physics modes of the Virtual Tri-HB.

Mode	Loading family	Topology / variant	True-triaxial
1	Gas-gun uniaxial SHPB	single X axis	no
2	Confinement-chamber SHPB	X axis, passive lateral p_c	partial
3	Gas-gun Tri-HB static–dynamic	X dynamic, Y, Z pre-stress	yes
4	EM programmable	single-axis one-sided	no
5	EM programmable	multi-axis one-sided (X, XY, XYZ)	yes
6	EM programmable	symmetric opposing ($\pm X \pm Y \pm Z$)	yes (hydrostatic)

are proportional, giving the Hopkinson identity

$$v(x,t) = -C_0\, \varepsilon(x,t), \tag{3}$$

which underlies all bar-gauge analysis: a resistance gauge that measures strain also measures the particle velocity through the bar wave speed.

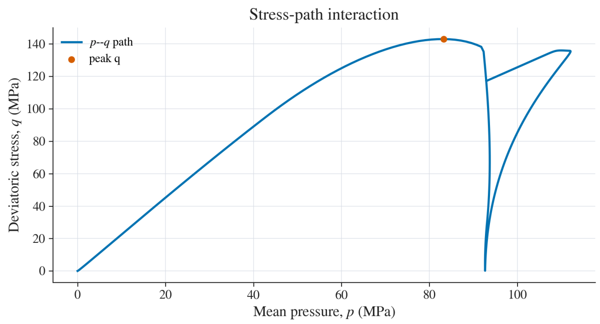
At a planar interface between media of acoustic impedance $Z = \rho c$, continuity of traction and velocity gives the stress reflection and transmission coefficients[1]

$$R = \frac{\sigma_R}{\sigma_I} = \frac{Z_2 - Z_1}{Z_1 + Z_2}, \qquad T = \frac{\sigma_T}{\sigma_I} = \frac{2Z_2}{Z_1 + Z_2}. \tag{4}$$

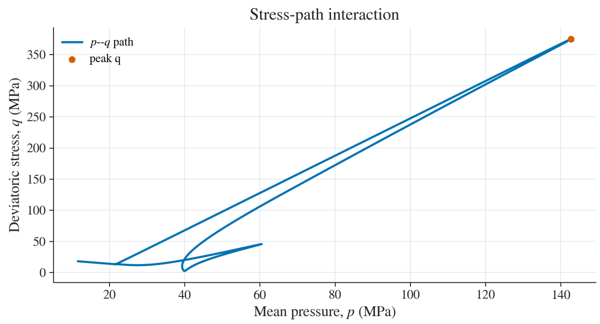
For a rock specimen softer than the steel bars ($Z_s < Z_b$) the reflection coefficient is negative, so the reflected pulse is tensile — the origin of the negative reflected strain $\varepsilon_{\mathrm{re}}$ recorded in compression tests, and a sign-diagnostic of the impedance match.

Equation (2) is exact only in the long-wavelength limit. The exact axisymmetric (Pochhammer–Chree) solution makes the phase velocity frequency-dependent, $c_\phi(ka) = C_0\,\Phi(ka, \nu_b)$, where k is the wavenumber and a the bar radius: $\Phi \approx 1$ for $ka \lesssim 0.1$ but falls by a few percent as $ka \to 1$, so an unshaped rectangular pulse — rich in high-ka harmonics — develops spurious oscillations as it travels. Dispersion is negligible while the rise time τ_r satisfies $a/(c_\phi \tau_r) \lesssim 0.1$. The Tri-HB therefore uses pulse-shaped (gas-gun) and half-sine (electromagnetic) incident waves, which concentrate their energy at $ka \ll 1$ and are essentially dispersion-free for the bar geometries used here; the digital twin represents every loading mode with a smooth half-sine incident wave for the same reason.

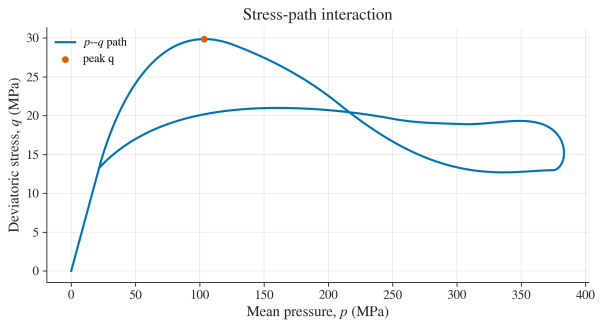
[1]For a 42CrMo steel bar $Z_b = \rho_b C_0 \approx 4.06 \times 10^7\,\mathrm{kg\,m^{-2}\,s^{-1}}$.



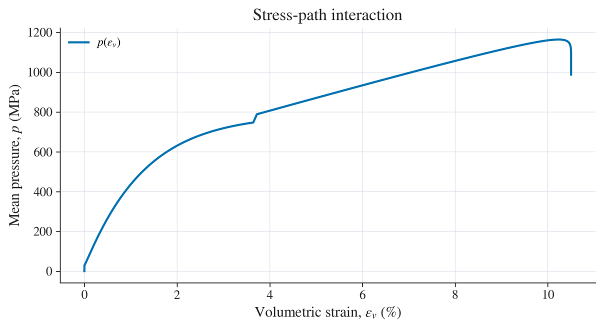
(a) Mode 1 — gas-gun uniaxial SHPB



(b) Mode 3 — gas-gun Tri-HB



(c) Mode 5 — EM asynchronous triaxial



(d) Mode 6 — EM symmetric hydrostatic

Figure 2: Mean-pressure–deviatoric-stress (p–q) trajectories for four of the six modes. Uniaxial loading (a) traces a steep deviatoric path; the triaxial modes (b, c) populate the multiaxial p–q plane, and the symmetric hydrostatic mode (d) collapses onto the $q \approx 0$ axis. The spread of accessible stress paths is the analytical advantage of the Tri-HB.

3.2 Force–velocity continuity and the three-wave method

At the incident-bar/specimen face the bar carries the superposed incident and reflected waves, while the transmission bar carries only the outgoing wave, so the two face forces and velocities are

$$F_1 = E_b A_b\,(\varepsilon_{\text{in}} + \varepsilon_{\text{re}}), \quad v_1 = C_0(\varepsilon_{\text{in}} - \varepsilon_{\text{re}}); \qquad F_2 = E_b A_b\,\varepsilon_{\text{tr}}, \quad v_2 = C_0\,\varepsilon_{\text{tr}}. \tag{5}$$

Once the specimen has reverberated into dynamic stress equilibrium ($t \gtrsim 3\,t_{\text{travel}}$, $t_{\text{travel}} = L_s/c_p$), the average axial stress, strain rate and strain follow from (5) as the standard three-wave relations

$$\sigma_x(t) = \frac{F_1 + F_2}{2A_s} = \frac{E_b A_b}{2A_s}\left[\varepsilon_{\text{in}} + \varepsilon_{\text{re}} + \varepsilon_{\text{tr}}\right] = \frac{E_b A_b}{A_s}\,\varepsilon_{\text{tr}}(t), \tag{6}$$

$$\dot\varepsilon(t) = \frac{v_1 - v_2}{L_s} = \frac{C_0}{L_s}\left[\varepsilon_{\text{in}} - \varepsilon_{\text{re}} - \varepsilon_{\text{tr}}\right] = -\frac{2C_0}{L_s}\,\varepsilon_{\text{re}}(t), \qquad \varepsilon(t) = \int_0^t \dot\varepsilon\,\mathrm{d}\tau, \tag{7}$$

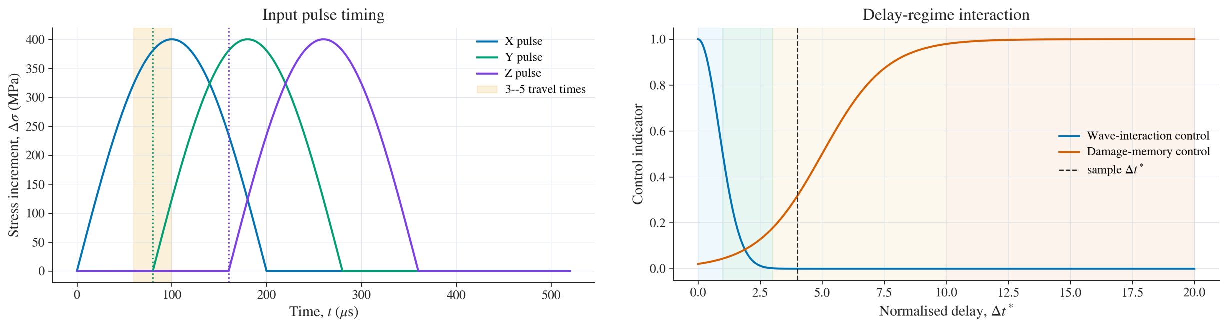
where A_b, A_s are the bar and specimen cross-sections and L_s the specimen length; the right-hand (one-wave) forms hold under perfect equilibrium $\varepsilon_{\text{in}} + \varepsilon_{\text{re}} = \varepsilon_{\text{tr}}$, and their agreement with the two-face forms is the reducer's built-in equilibrium check. Equilibrium itself is quantified by the force-balance ratio

$$\mathcal{R}(t) = \frac{|F_1(t) - F_2(t)|}{\frac{1}{2}(F_1(t) + F_2(t))} < 0.05, \qquad t_{\text{eq}} \approx (3\text{–}5)\,L_s/c_p. \tag{8}$$

The lateral response is obtained from the two output bars on each transverse axis,

$$\sigma_y(t) = \frac{E_b A_b}{2A_s}\left[\varepsilon_{y1} + \varepsilon_{y2}\right], \qquad \sigma_z(t) = \frac{E_b A_b}{2A_s}\left[\varepsilon_{z1} + \varepsilon_{z2}\right], \tag{9}$$

with the analogous strain integrals. These lateral outputs are Poisson-driven: the X pulse compresses the specimen, which expands laterally and drives output waves outward along the Y



(a) Per-axis pulse timing (b) Delay-regime interaction map

Figure 3: Wave-interaction analysis (Step 2). (a) Independently delayed X, Y, Z pulses with the 3–5 travel-time equilibrium window shaded. (b) The normalised-delay regime map separating superposition-controlled from damage-memory-controlled response.

and Z bars, recorded by gauges G3–G6. The Virtual Tri-HB performs this reduction simultaneously on all three axes, so a single test yields the full $\sigma_x, \sigma_y, \sigma_z$ history rather than one axial curve.

3.3 Validity of the three-wave method under static pre-stress

In the static–dynamic configuration the hydraulic cylinders pre-load the bars before the pulse arrives, so the bar ends are no longer free. Because the bar is linear-elastic, the total field superposes a uniform static part and the propagating dynamic part, $\sigma = \sigma_s + \sigma_d$ with $\sigma_s = F_{\text{pre}}/A_b = \text{const}$ ($\mathrm{d}\sigma_s/\mathrm{d}x = 0$). Substituting into (2), the static part drops out and the dynamic part obeys the same wave equation, so the three-wave relations apply unchanged to the dynamic *increments* $\varepsilon_{\text{in}}, \varepsilon_{\text{re}}, \varepsilon_{\text{tr}}$ — the pre-stress is merely a constant gauge offset, removed by baselining before the pulse. The superposition is valid provided the bar stays elastic at all times,

$$\sigma_{d,\text{peak}} < \sigma_{\text{prop}} - |\sigma_s|, \tag{10}$$

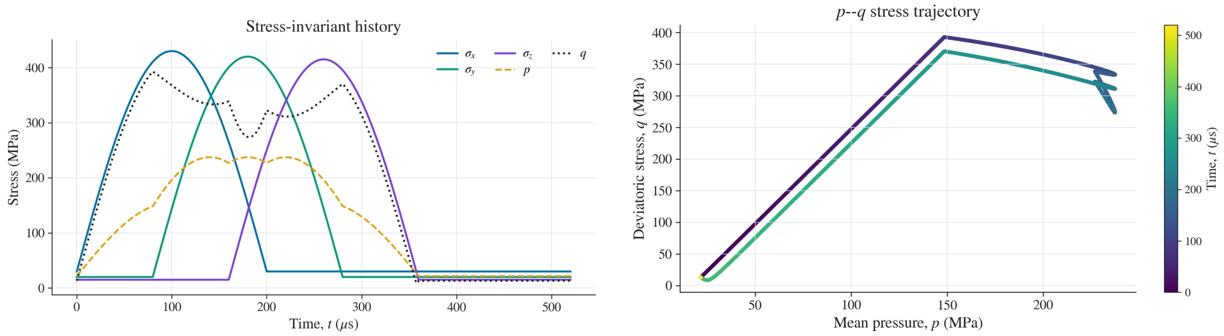
so the pre-stress consumes part of the dynamic-amplitude budget (the bar-plasticity limit the workspace enforces); the wave speed $C_0 = \sqrt{E_b/\rho_b}$ is unchanged to within a sub-percent acoustoelastic shift, and the quasi-static actuator leaves the end's *dynamic* impedance — and hence the reflection seen by the pulse — effectively that of a free bar over the μs event. This superposition is what legitimises static–dynamic coupled testing.

3.4 Wave-interaction regime map

For multi-axis loading the relative timing of the per-axis pulses controls whether the specimen responds by wave superposition or by sequential damage-memory. Defining a normalised inter-axis delay $\Delta t^* = \Delta t / t_{\text{travel}}$, with $t_{\text{travel}} = L_s/c_p$ the specimen transit time, the workflow classifies the response into synchronous superposition ($\Delta t^* < 1$), reverberation-coupled ($1 \le \Delta t^* \le 3$), transitional ($3 < \Delta t^* \le 10$) and sequential damage-memory ($\Delta t^* > 10$) regimes (Fig. 3). This map links the *timing* of triaxial loading to the *mechanism* of failure, a relationship unique to programmable multiaxial systems.

4 The integrated digital twin

The Virtual Tri-HB (`tri_hb_integrated.py`) is a single application that bundles six analysis steps under one shared experimental setup: (1) configuration and either simulation or reduction of bar-gauge data; (2) wave-propagation and equilibrium analysis; (3) stress-path and invariant



(a) Stress and invariant histories (b) p–q trajectory (time-coloured)

Figure 4: Stress-path analysis (Step 3) for the reference asynchronous-XYZ case: total principal stresses with the mean and deviatoric invariants (a), and the resulting p–q trajectory traced in time (b).

analysis; (4) damage and energy analysis with directional descriptors; (5) a publication-ready summary; and (6) automatic generation of a report and presentation (LaTeX + PDF) from the current run. A separate page renders the live calibration/validation plan. Each step inherits the shared configuration, so the wave, stress-path and damage views remain mutually consistent. The stress-path stage resolves the total stresses into the invariants

$$p = \tfrac{1}{3}(\sigma_x + \sigma_y + \sigma_z), \quad q = \sqrt{\tfrac{1}{2}\left[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2\right]}, \quad \cos 3\theta = \frac{3\sqrt{3}}{2}\frac{J_3}{J_2^{3/2}}, \tag{11}$$

so that all three principal stresses — including the intermediate stress through the Lode angle θ — enter the subsequent failure analysis (Fig. 4).

5 Constitutive advances

5.1 True-triaxial, rate-dependent failure envelope

Failure is evaluated through the index $F = q/q_f$ against a pressure-dependent envelope with a deviatoric (Lode) shape and an optional rate dependence,

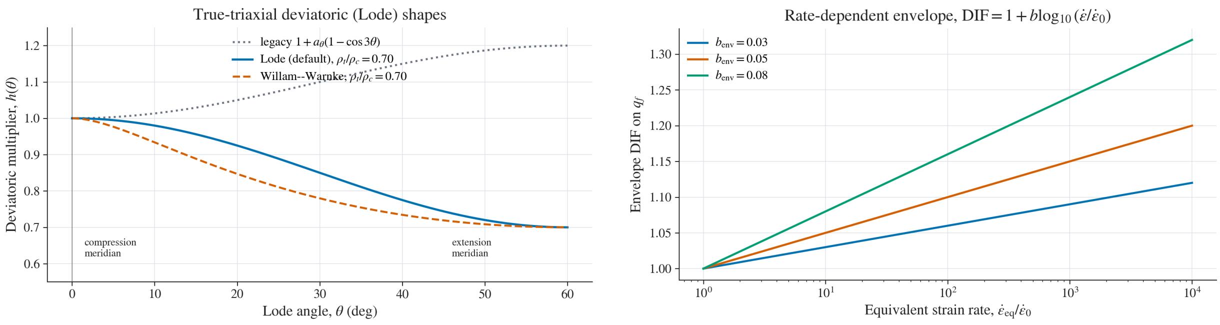
$$q_f = (A_{\text{eff}} + B_{\text{eff}}\, p^n)\, h(\theta), \qquad A_{\text{eff}} = A\,\text{DIF}, \quad \text{DIF} = 1 + b_{\text{env}} \log_{10}\max\left(\tfrac{|\dot\varepsilon_{\text{eq}}|}{\dot\varepsilon_0}, 1\right). \tag{12}$$

The meridian intercept and slope A, B are derived consistently from the unconfined strength and confinement ratio, ensuring $F = 1$ at the static failure point. The deviatoric shape $h(\theta)$ admits three forms: a legacy cap, a linear extension-weakened interpolation, and a convex Willam–Warnke form (Willam and Warnke, 1975). Both new forms use the meridian ratio ρ_t/ρ_c and make the extension meridian *weaker* than compression — the physically correct sense for brittle rock — so the intermediate principal stress effect enters with the right sign (Fig. 5a). The default is the Lode form with $\rho_t/\rho_c = 0.70$. The optional rate term in Eq. (12) inflates the envelope with strain rate (Fig. 5b), so that the *onset* of failure shifts dynamically, complementing the rate sensitivity of the damage kinetics (Zhao, 2000).

5.2 Thermodynamically consistent damage dissipation

Damage D evolves by a Perzyna-type overstress law with load-shedding feedback,

$$\dot D = \frac{(1-D)^\alpha}{\tau_D}\left\langle \frac{F_{\text{eff}} - 1}{F_0}\right\rangle^m \left(\frac{|\dot\varepsilon_{\text{eq}}|}{\dot\varepsilon_0}\right)^\beta, \qquad F_{\text{eff}} = (1-D)\,F, \tag{13}$$



(a) Deviatoric (Lode) shapes (b) Envelope dynamic increase factor

Figure 5: True-triaxial and rate-dependent failure envelope. (a) The default Lode and Willam–Warnke shapes correctly weaken the extension meridian relative to compression ($\rho_t/\rho_c = 0.70$), capturing the σ_2 effect; the legacy cap (dotted) has the opposite, unphysical sense. (b) Optional envelope DIF versus normalised strain rate.

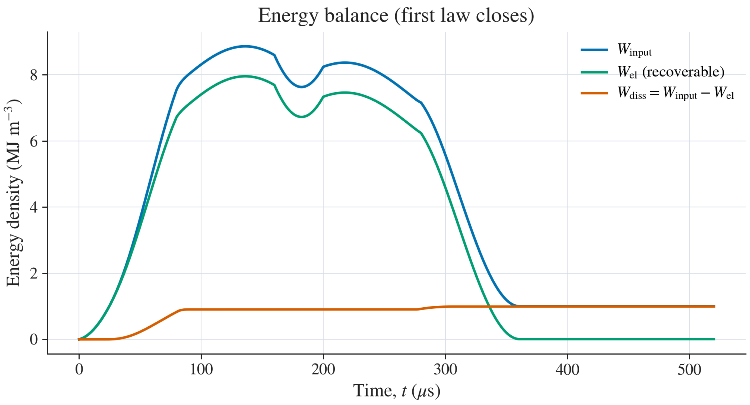


Figure 6: Energy-density partition (Step 4). With the damage-coupled strain the balance closes exactly: $W_{\text{input}} = W_{\text{el}} + W_{\text{diss}}$, with W_{diss} the non-recoverable (dissipated) energy.

so that D self-limits rather than snapping to unity (Perzyna, 1966; Lemaitre, 1985). Crucially, the energy partition is now derived from a single constitutive relation. Damage degrades the secant modulus to $E_0(1 - D)$, so the specimen carries the damage-coupled elastic strain and the work, recoverable energy and dissipation become

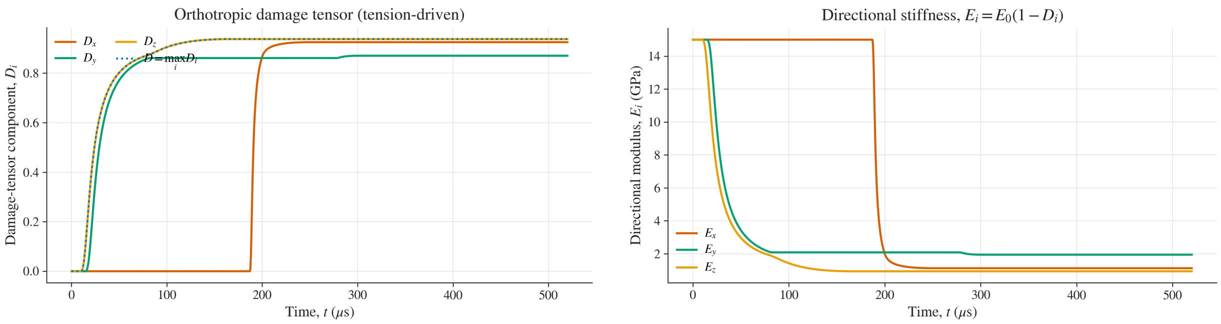
$$\varepsilon_i = \frac{\sigma_i - \nu(\sigma_j + \sigma_k)}{E_0(1-D)}, \quad W_{\text{input}} = \int_0^t \sigma_i \dot{\varepsilon}_i \, \mathrm{d}\tau, \quad W_{\text{el}} = \tfrac{1}{2}\sigma_i\varepsilon_i, \quad W_{\text{diss}} = W_{\text{input}} - W_{\text{el}} \geq 0. \tag{14}$$

The first law then closes by construction, $W_{\text{input}} = W_{\text{el}} + W_{\text{diss}}$: the input work ends above zero by exactly the dissipated part, the recoverable energy returns toward zero on unloading, and the dissipation rises monotonically (Fig. 6). This removes the inconsistency of treating the input work as fully recoverable while estimating dissipation from a separate integral.

5.3 Orthotropic damage for blasting-induced fracturing

Rock blasting produces *directional* fracture: radial cracks and spall open perpendicular to the local tension, so the stiffness and wave-speed loss differ by direction. A scalar D cannot represent this. The Tri-HB digital twin therefore offers a diagonal damage tensor $\mathbf{D} = \mathrm{diag}(D_x, D_y, D_z)$ aligned with the bars, each component driven by the directional tensile (extensile) strain through the same overstress kinetics,

$$\dot{D}_i = \frac{(1-D_i)^\alpha}{\tau_D} \Big\langle \frac{(1-D_i)F_i - 1}{F_0} \Big\rangle^m \Big(\frac{|\dot{\varepsilon}_{\text{eq}}|}{\dot{\varepsilon}_0}\Big)^\beta, \qquad F_i = \frac{\langle -\varepsilon_i \rangle_+}{\varepsilon_{t0}}, \quad \varepsilon_{t0} = \frac{\sigma_t}{E_0}, \tag{15}$$



(a) Damage-tensor components D_x, D_y, D_z (b) Directional stiffness $E_i = E_0(1 - D_i)$

Figure 7: Orthotropic (Stage-1) damage for the reference triaxial case. (a) The diagonal damage-tensor components, each grown by its own directional tensile strain; the scalar $D = \max_i D_i$ governs the energy/stiffness summary. (b) The resulting directional modulus loss, the per-axis quantity the three bar pairs measure.

for $i = x, y, z$. Damage grows only under directional extension — there is no damage in pure compression — so axial X compression drives the lateral components D_y, D_z (axial splitting) while $D_x \approx 0$, and confinement on an axis suppresses its damage (Fig. 7a). The per-axis modulus degrades independently, $E_i = E_0(1 - D_i)$ (Fig. 7b), as does the directional wave speed $c_{p,i} = c_{p0}\sqrt{1 - D_i}$. This formulation is rooted in the microcrack-damage tradition for brittle rock (Grady and Kipp, 1980; Taylor et al., 1986) but resolves damage by direction. Decisively, because the three orthogonal bars measure direction-resolved modulus and wave-speed loss, each D_i can be calibrated *component-by-component* from a single true-triaxial test — a measurement a uniaxial SHPB cannot provide.

6 Validation and calibration roadmap

The advances above are constitutive and computational; the system is presented here as a measurement-and-modelling framework rather than a validated predictive model. The decisive step is experimental: a staged, hierarchical calibration that fixes the elastic and strength parameters before the damage kinetics, on a *calibration set* of tests, followed by *blind prediction* on a disjoint *validation set*. The Tri-HB's unique strength is that its three independent axes allow each anisotropic-damage component, each meridian of the failure surface and each rate level to be probed directly. A practical matrix calibrates the meridian and Lode parameters on Modes 1–3 and blind-predicts the EM multiaxial Modes 4–5, while the directional damage components D_x, D_y, D_z are recovered post-test from per-axis ultrasonic modulus loss, $D_i = 1 - E_{i,\mathrm{post}}/E_0$. Demonstrating that one calibrated parameter set predicts strength and directional fracture on a held-out loading path is what converts the framework from phenomenological to predictive, and is the natural subject of the experimental campaign now enabled by the system.

7 Conclusions

We have presented the Monash true-triaxial Hopkinson bar and its integrated digital twin. The principal advances are: (i) a unified four-family, six-mode treatment that spans uniaxial SHPB to programmable multiaxial loading on a single shared setup; (ii) a true-triaxial, Lode-angle- and rate-dependent failure envelope that captures the intermediate principal stress effect with the correct sign; (iii) a thermodynamically consistent damage-energy partition in which the first law closes by construction; and (iv) an orthotropic damage tensor whose directional components reproduce the oriented fracture of rock blasting and can be calibrated component-by-component from the system's direction-resolved measurements. Together, the hardware and its digital twin establish a platform for quantitative, multiaxial dynamic constitutive modelling of rock; the immediate next step is the experimental calibrate-then-blind-predict campaign that the system uniquely enables.

Declarations

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Competing interests. The author declares no competing interests.

Data and code availability. The Virtual Tri-HB workspace (`tri_hb_integrated.py`) and the figure-generation scripts used here are part of the Tri-HB repository. All figures in this paper are produced from prescribed-pulse reference cases by the workspace's own analysis pipeline; no experimental data were used.

Author contributions. Q.B.Z. conceived the system and the digital-twin framework, implemented the analysis pipeline, and wrote the manuscript.

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